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(54) **WEARABLE HEALTH MONITORING SYSTEM AND METHOD USING HUMAN BODY COMMUNICATION**

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H04B 1/10	(2006.01)
H04W 72/08	(2009.01)
G06F 21/32	(2013.01)

(52) **U.S. Cl.**

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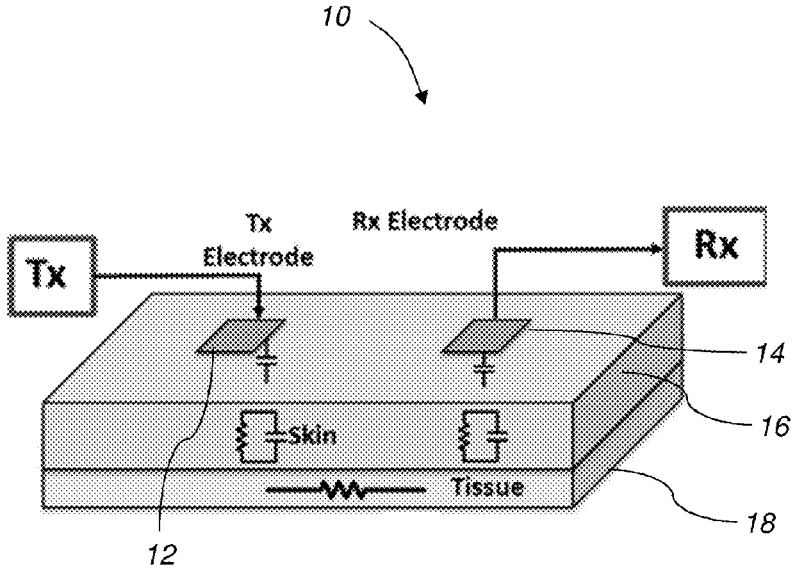
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(57) **ABSTRACT**

A communication interference rejection system comprising a relatively low-impedance voltage mode driver output which receives a signal from a sensor, the sensor near a user's skin, a receiver operatively connected to a device connected to a body of a user, wherein the receiver configured to receive a signal transmitted through the body of the user. The signal comprises a relatively substantially small constant amplitude component and a relatively large sinusoidal or modulated interference component, the interference component due to human body antenna effect, the receiver comprising a high-impedance termination to minimize channel loss, the receiver further configured to receive frequencies in the 10 KHz to 10 MHz range.

4 Claims, 7 Drawing Sheets



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CPC *G06F 21/32* (2013.01); *H04B 1/1036* (2013.01); *H04W 72/082* (2013.01)

(58) **Field of Classification Search**
CPC .. A61B 5/0028; H04B 1/1036; H04W 72/082
See application file for complete search history.

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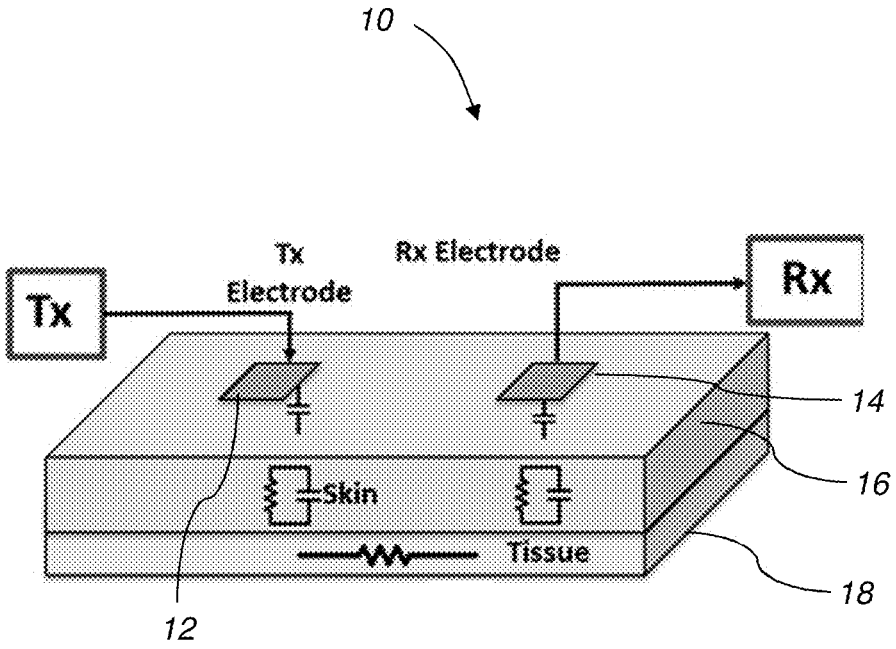


FIG. 1A

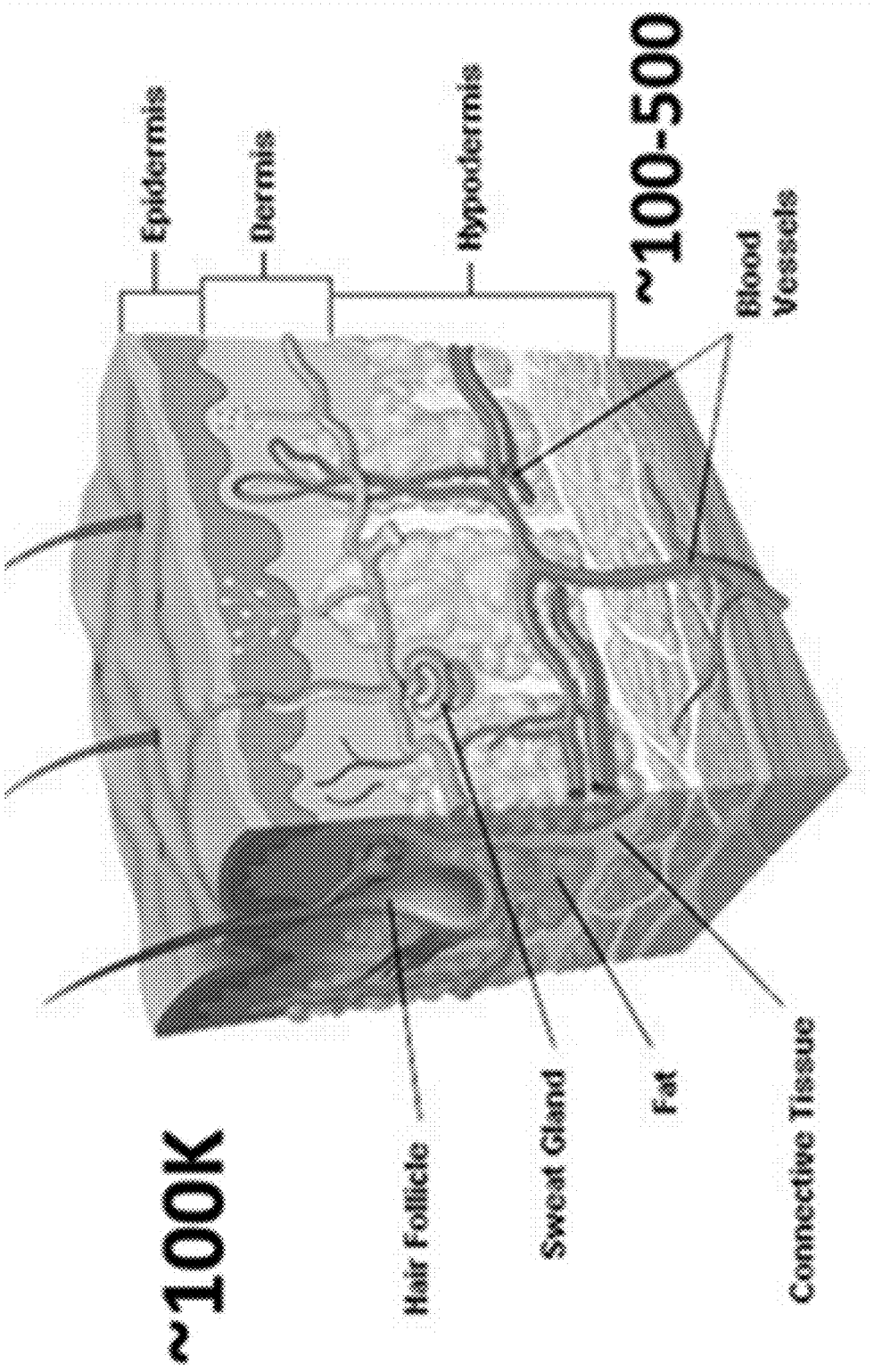

Epidermis
Dermis
Hypodermis
~100-500
Blood Vessels
~100K
Hair Follicle
Sweat Gland
Fat
Connective Tissue

FIG. 1B

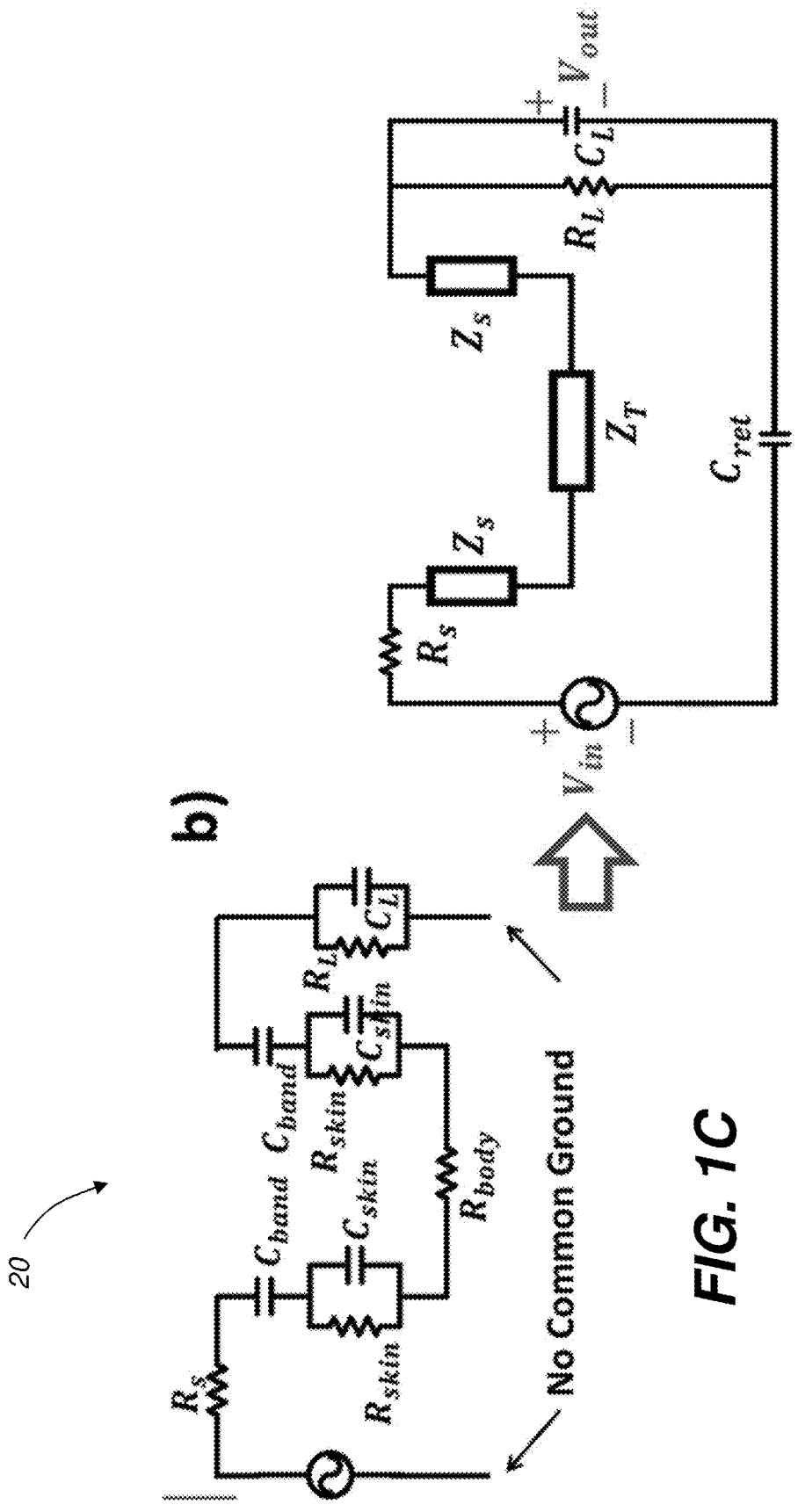

20
b)
R_S
R_{skin}
C_{band}
C_{band}
C_{skin}
R_{skin}
R_{body}
C_L
R_L
C_{skin}
No Common Ground
V_{in}
R_S
Z_S
Z_T
Z_S
C_{ret}
R_L
C_L
V_{out}

FIG. 1C

FIG. 1D

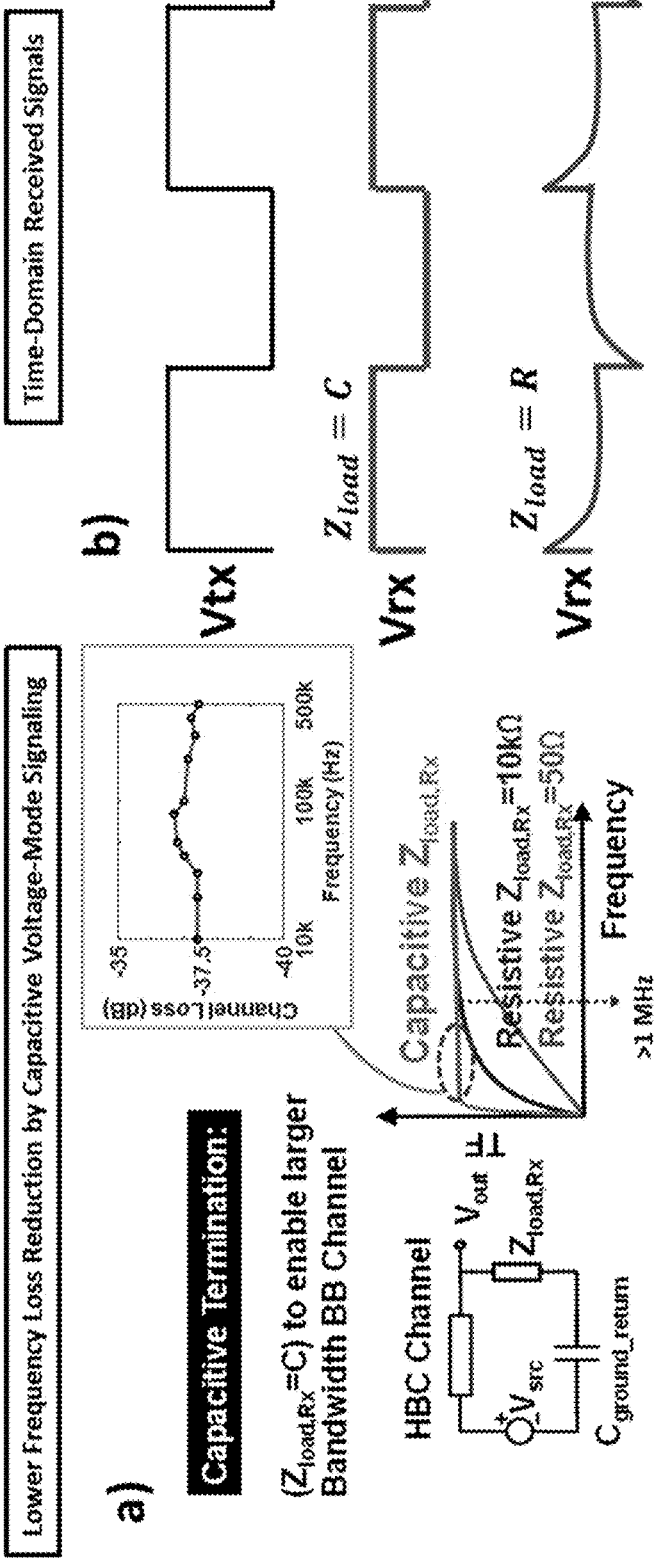

Lower Frequency Loss Reduction by Capacitive Voltage-Mode Signaling
a)
Capacitive Termination:
($Z_{load,Rx}$=C) to enable larger Bandwidth BB Channel
HBC Channel
V_{out}
$Z_{load,Rx}$
V_{src}
C_{ground_return}
Channel Loss (dB)
-35
-37.5
-40
10k
100k
500k
Frequency (Hz)
Capacitive $Z_{load,Rx}$
Resistive $Z_{load,Rx}$=10kΩ
Resistive $Z_{load,Rx}$=50Ω
Frequency
>1 MHz
b)
Time-Domain Received Signals
Vtx
time
$Z_{load} = C$
Vrx
time
$Z_{load} = R$
Vrx
time

FIG. 2A

FIG. 2B

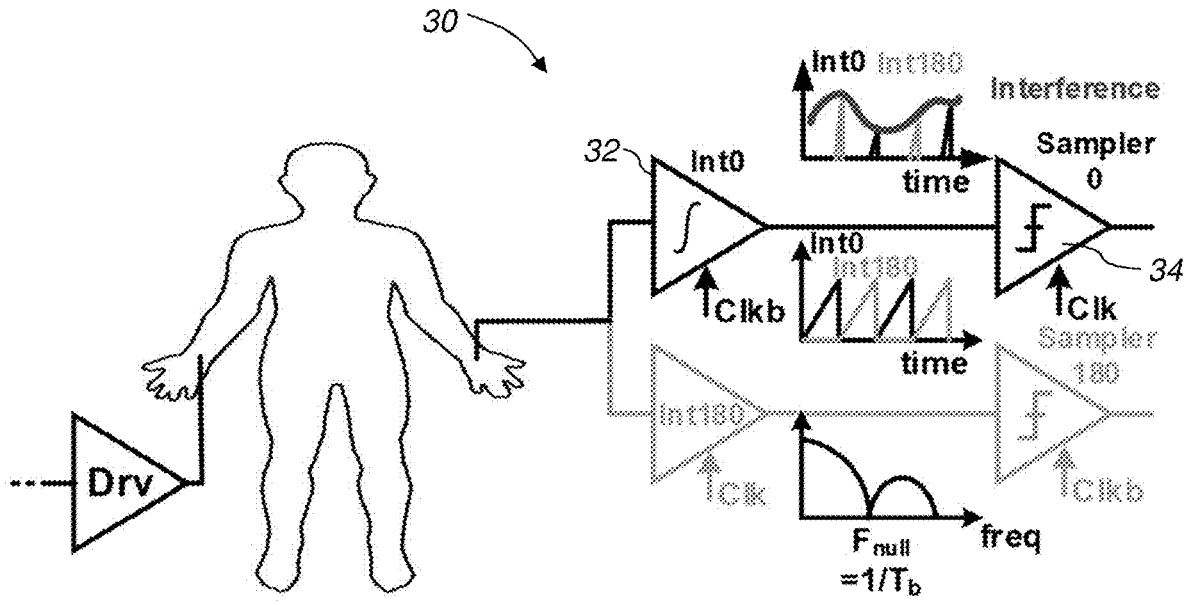

30
Int0
Int180
Interference
Sampler
0
32
Int0
time
34
Clkb
Clk
Int0
Int180
time
Sampler
180
Int180
Clk
Clkb
Drv
Fnull
=1/Tb
freq

FIG. 3A

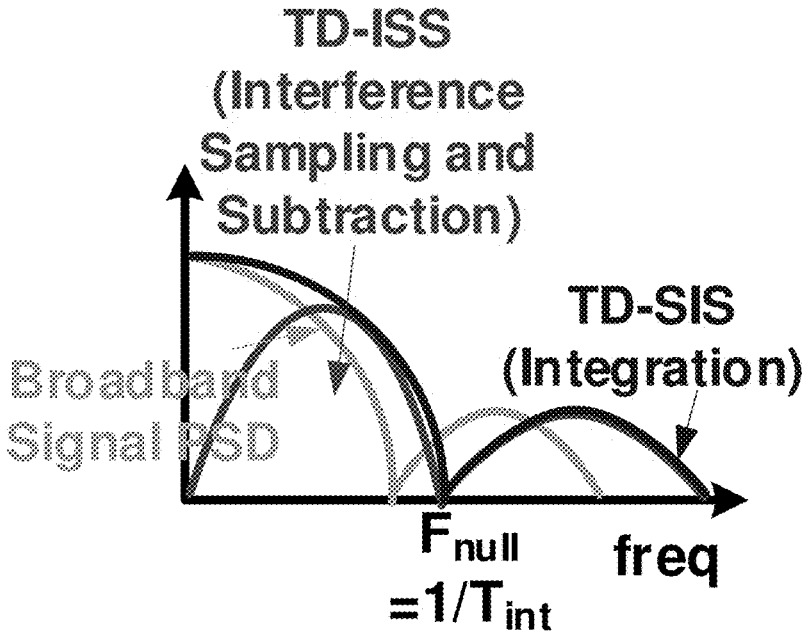

TD-ISS
(Interference
Sampling and
Subtraction)
TD-SIS
(Integration)
Broadband
Signal PSD
Fnull
=1/Tint
freq

FIG. 3B

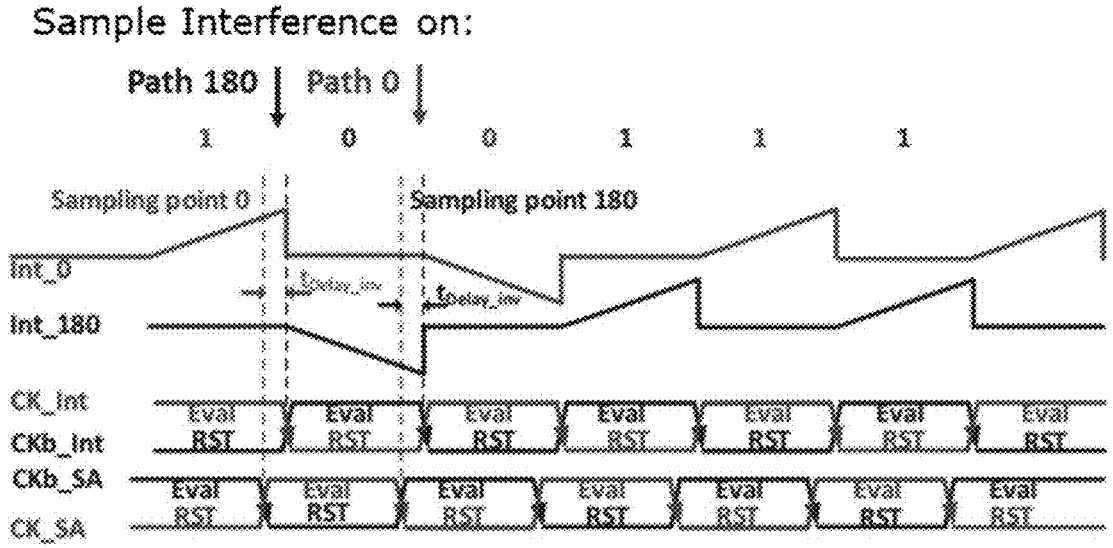


FIG. 3C

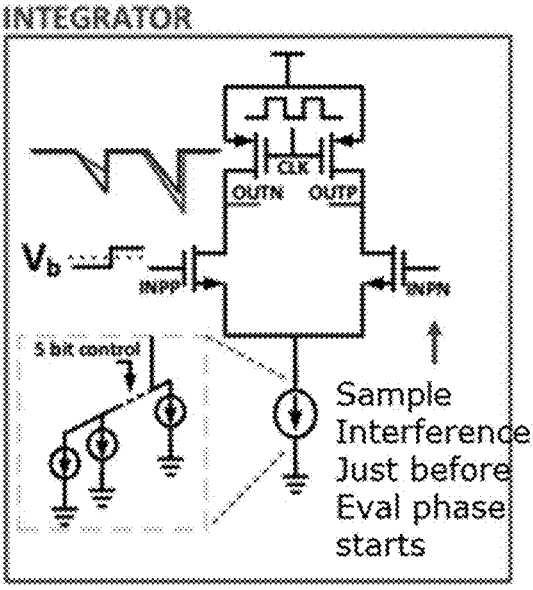


FIG. 3D

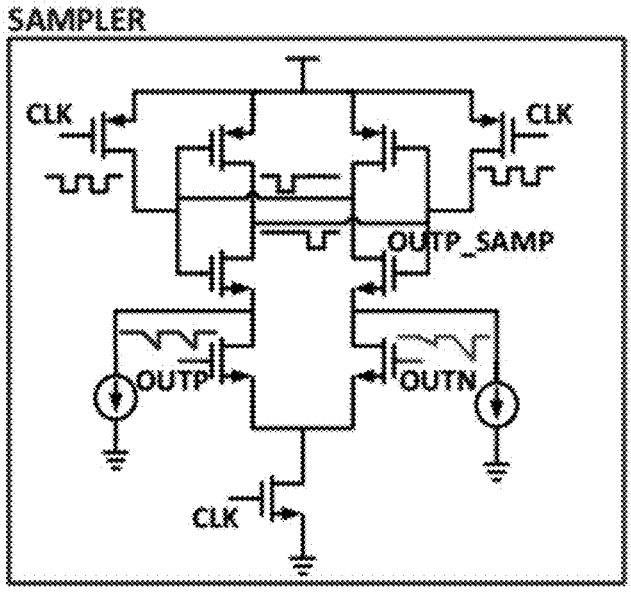


FIG. 3E

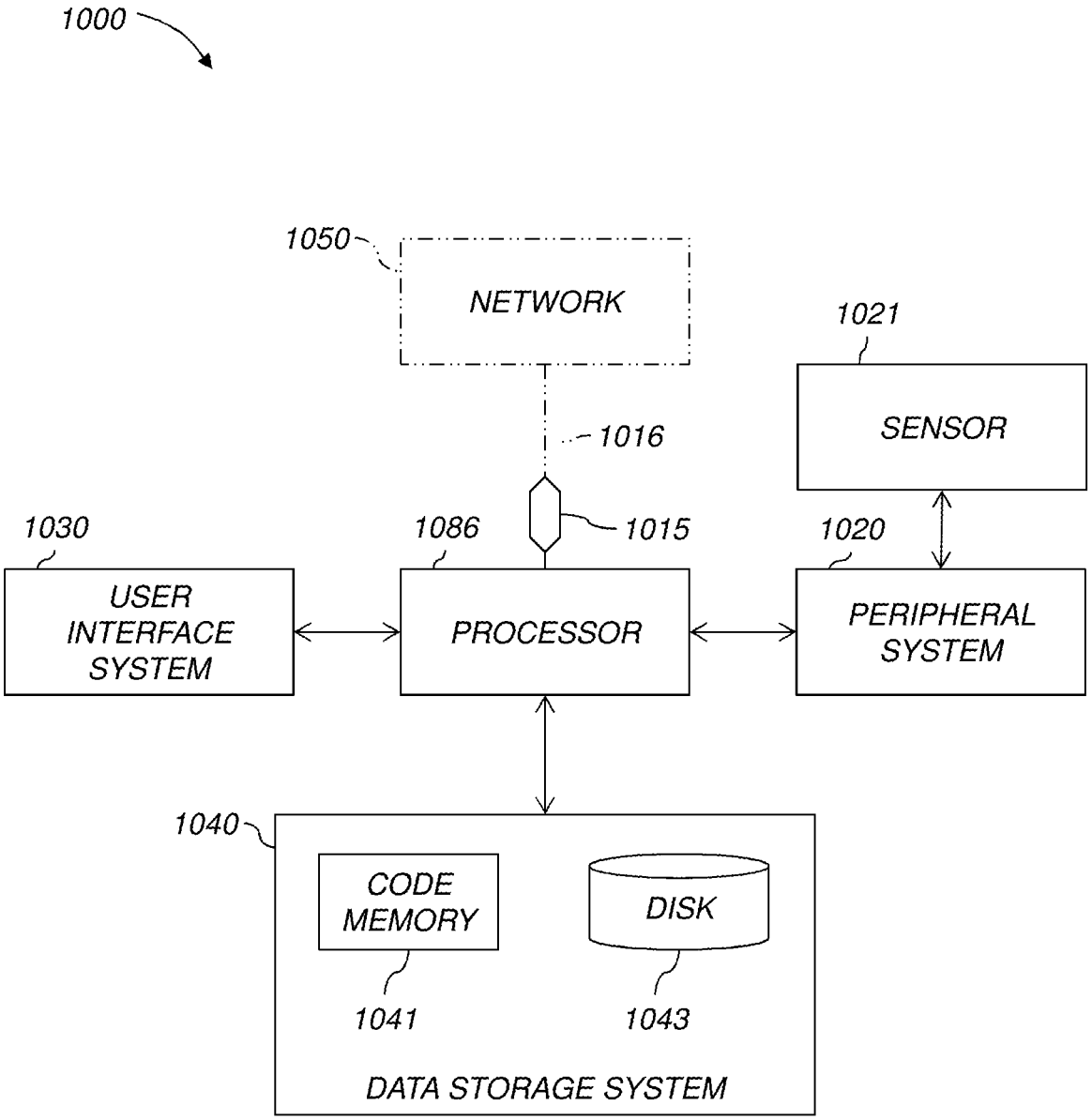

1000
1050
NETWORK
1021
SENSOR
1016
1030
1086
1015
1020
USER INTERFACE SYSTEM
PROCESSOR
PERIPHERAL SYSTEM
1040
CODE MEMORY
DISK
1041
1043
DATA STORAGE SYSTEM

FIG. 4

WEARABLE HEALTH MONITORING SYSTEM AND METHOD USING HUMAN BODY COMMUNICATION

CROSS REFERENCE TO RELATED APPLICATIONS

The present patent application relates to and claims the priority benefit of U.S. Provisional Application Ser. No. 62/671,102, filed May 14, 2018, the content of which is hereby incorporated herein by reference in its entirety.

TECHNICAL FIELD

The present disclosure is related to electronic communication systems, and more specifically, to electronic authentication systems which use the human body as part of a signal transmission path.

BACKGROUND

The continuous reduction of size of unit computing, has propelled the growth of wearable sensors and computing devices (e.g. Fitness trackers, Smart watches). Soon, the human body will become a platform for interconnected wearable smart devices, which will aid and improve human quality of life. This calls for efficient ways to connect these wearable devices on the human body. Moreover, since each individual will be wearing a large amount of information on their body (i.e., the Human Intranet), they can now transmit this information to other humans or machines (Human Internet) at their will or use this information for secure authentication. Such on-body wearable devices are typically interconnected using a wireless body area network (WBAN). Human Body Communication (HBC) has recently emerged as a strong contender for this human body network, as it provides ultra-low power (ULP) and increased security, compared to WBAN. ULP is achieved as human body is used as a conducting medium, which exhibits significantly lower loss than radio frequency propagation through air. HBC is more secure as the information is contained within the human body and cannot be snooped on unless the person is physically touched, unlike WBAN, where the wireless signals can be easily snooped on by an attacker.

One disadvantage of HBC is that the human body acts as an antenna at the FM frequency band. This has been the biggest bottleneck in high-speed ULP HBC implementation. Signaling techniques that allow circumvention of the interference, such as adaptive frequency hopping (AFH) and fixed narrowband signaling have been proposed. However, such systems provide no way to suppress the interference other than avoiding it using adaptive/fixed narrowband signaling, which leads to energy-inefficient implementation and requires bulky filters. Therefore, improvements are needed in the field.

BRIEF DESCRIPTION OF THE DRAWINGS

The above and other objects, features, and advantages of the present invention will become more apparent when taken in conjunction with the following description and drawings wherein identical reference numerals have been used, where possible, to designate identical features that are common to the figures, and wherein:

FIG. **1**A shows a diagram of a human body communication system according to one embodiment.

FIG. **1**B shows a sample diagram of human skin and associated tissue.

FIG. **1**C shows a schematic diagram of the human body communication system of FIG. **1**A according to one embodiment.

FIG. **1**D shows a schematic diagram of a lumped model of the HBC system of FIG. **1***a* according to one embodiment.

FIG. **2**A shows an HBC system which employs capacitive voltage-mode signaling according to one embodiment.

FIG. **2**B shows the associated time-domain received signals for the system of FIG. **2***a*.

FIG. **3**A shows an HBC system which employs a time-domain interference sampling and subtraction technique according to one embodiment.

FIG. **3**B shows a comparison of gain vs. frequency plots for various methods.

FIG. **3**C shows the associated time domain signals for the system of FIG. **3**A according to one embodiment.

FIG. **3**D shows the integrator circuit portion of the HBC system of FIG. **3**A.

FIG. **3**E shows the sampler circuit portion of the HBC system of FIG. **3**A.

FIG. **4** shows a diagram of a data-processing system having human body communication functionality according to one embodiment.

DETAILED DESCRIPTION

Throughout this description, some aspects are described in terms that would ordinarily be implemented as software programs. Those skilled in the art will readily recognize that the equivalent of such software can also be constructed in hardware, firmware, or micro-code. Because data-manipulation algorithms and systems are well known, the present description is directed in particular to algorithms and systems forming part of, or cooperating more directly with, systems and methods described herein. Other aspects of such algorithms and systems, and hardware or software for producing and otherwise processing signals or data involved therewith, not specifically shown or described herein, are selected from such systems, algorithms, components, and elements known in the art. Given the systems and methods as described herein, software not specifically shown, suggested, or described herein that is useful for implementation of any aspect is conventional and within the ordinary skill in such arts.

Various aspects relate to communication systems utilizing the human body as an electrical signal pathway.

The present disclosure provides an electronic communication system which utilizes capacitive voltage-mode communication or time-domain interference sampling between wearable health monitoring devices, which enables low energy, secure communication necessary for such energy constrained systems.

In view of the foregoing, various aspects provide improved reliability of an electronic communication system. A technical effect is to electronically communicate data between wearable health monitoring devices connected to the human body.

The human body can be used as a broadband (BB) channel for data transmission. A broadband channel with ~MHz bandwidth can enable data transmission at megabits/second speed which is sufficient for applications such as image or data transfer. FIG. **1**A shows a diagram of a HBC system **10** having transmit electrode **12** and receive electrode **14** adjacent or slightly separated from human skin **16**, with associated lower tissue **18**. FIG. **1**B shows a typical

human skin and tissue anatomy. FIG. **1**C shows a detailed circuit diagram of a circuit **20** for capacitive HBC between two wearable devices. The circuit model **20** shows there is no common ground, meaning no closed path for current to flow. The closed path is formed using parasitic capacitances from device **1** and **2** to earth's ground (Cret), leading to the name capacitive HBC. FIG. **1**D shows a simplified lumped model of a capacitive HBC between two wearable devices. Z_S is a combination of the skin-electrode contact resistance and the skin tissue resistance and is in the order of KΩ. Once the signal is coupled to the body it is transferred through the low resistance tissue layers inside the body, whose impedance is modeled as Z_T in FIG. **1***b*. The source impedance (R_S) is of the order of few ohms. The load could consist of a parallel combination of resistance (R_L), capacitance (C_L).

Voltage-Mode Signaling (High-Impedance Termination): Since the summation of skin and tissue impedance is above the KΩ range, a 50Ω or other low-impedance termination at the receiver creates an unfavorable voltage division, leading to a high-loss at low frequencies (FIG. **2**A). The load resistance should be chosen as a high-impedance on the order of MOs (e.g. 1-10 MΩs, 1-100 MΩs or 1-900 MΩs). Efficient communication in such a scenario is achieved by focusing on the voltage transfer characteristics of the human body channel. To that end, according to one embodiment, Voltage Mode (VM) Signaling is used for the communication between the devices in the HBC system (e.g., between the sensor and a worn hub unit, or between sensors). In VM signaling, the measured metric at different points of the system is voltage and voltage transmission is achieved by a low output impedance source and a high input impedance load. In one embodiment, a signal generator is used as a low impedance source and an oscilloscope is used as a high impedance load for measurements, however, the high impedance load may comprise other devices as well.

Capacitive Voltage-Mode Signaling: FIG. **2**A shows a Circuit diagram of capacitive HBC and example frequency response for capacitive termination and two cases of resistive termination. Capacitive termination allows close to flat-band response, making it suitable for broadband signaling. The closed path is formed using parasitic capacitances from device **1** and **2** to earth's ground (Cret), leading to the name capacitive HBC. Capacitive termination (C_L) is used at the receiver end and provides a flat-band frequency response as shown by the loss equation below:

$$\text{Channel Loss} = \frac{V_{rx}}{V_{tx}} = \frac{C_{ret}}{(C_{ret} + C_L)}$$

Since, the Channel Loss is not a function of frequency, it provides a flat-band frequency response, suitable for broadband signaling, that has energy in a broad bandwidth from DC to operating frequency. The return path is formed by a capacitance between the transmitter and the receiver. This return path capacitance and the capacitance at the receiver end forms a capacitive voltage division, as can be seen in FIG. **2**A. Since a capacitive voltage division ratio is independent of frequency, this results in a channel response with almost constant loss across all frequencies, resulting in a broadband channel. A broadband channel enables transmission of a signal through the human body directly as 1/0 bits without the need of any modulation or demodulation techniques. Since the broadband HBC enables maximum utilization of available bandwidth, systems utilizing the herein described HBC system will be more power-efficient than previous narrowband wireless or HBC based systems. The time-domain response to an example broadband data stream is shown in FIG. **2**B for capacitive termination and resistive termination HBC, showing the resistive termination suffers from quick signal loss with time, whereas the capacitive termination has a cleaner received signal.

Time-Domain Interference Sampling: The human body acts as an antenna and picks up interferences, which makes broadband communication through the body hard. An integrating DDR (I-DDR) receiver can handle interferences with frequency

$f_{intf}=nf_{data}$, where n is an positive integer

Any other frequencies $f_{intf}>f_{data}$ can be rejected by dynamically adjusting the integration time $T_{int}=n*T_{intf}$, which shows that this technique works well for interferences higher than data-rate, but does not provide much benefit for interference frequencies below data-rate.

Interference frequencies below data-rate can be rejected using the following Time-Domain Interference Sampling and Subtraction (TD-ISS) technique.

The details of the technique are shown in FIGS. **3**A-**3**E. FIG. **3**A shows an I-DDR receiver **30** used for interference-robust HBC signaling according to one embodiment, and the associated freq. vs. gain plot (FIG. **3**B) for the two methods (TD-ISS and TD-SIS). The interference is sampled and subtracted. More specifically, the incoming signal is integrated by the integrator **32**, then sampled by the sampler **34**. FIG. **3**C shows that the Sampling should be performed at the end of the Reset phase and the sampled value should be held on to the INPN of the integrator **32** (FIG. **3**D) during the evaluation phase, leading to a subtraction of the present signal from the sampled interference at the start of the integration period. If $f_{intf}\ll f_{data}$, it can be assumed that this sampled value is constant over the bit period, canceling these interferences. FIG. **3**E shows the circuit details of the sampler **34**.

This way, TD-ISS provides robustness to interferences $f_{intf}<f_{data}$ (interference sampling and subtraction) and $f_{intf}>f_{data}$ (integration) simultaneously.

Steps of various methods described herein can be performed in any order except when otherwise specified, or when data from an earlier step is used in a later step. Exemplary method(s) described herein are not limited to being carried out by components particularly identified in discussions of those methods.

FIG. **4** is a high-level diagram showing the components of an exemplary data-processing system **1000** for analyzing data and performing other analyses described herein, and related components. The system includes a processor **1086**, a peripheral system **1020**, a user interface system **1030**, and a data storage system **1040**. The peripheral system **1020**, the user interface system **1030** and the data storage system **1040** are communicatively connected to the processor **1086**. Processor **1086** can be communicatively connected to network **1050** (shown in phantom), e.g., the Internet or a leased line, as discussed below. The HBC data described herein may be received or sent through a human body via sensors **1021** (or similar transmitters) and/or displayed using display units (included in user interface system **1030**) which can each include one or more of systems **1086**, **1020**, **1030**, **1040**, and can each connect to one or more network(s) **1050**. Processor **1086**, and other processing devices described herein, can each include one or more microprocessors, microcontrollers, field-programmable gate arrays (FPGAs), application-specific integrated circuits (ASICs), programmable logic devices (PLDs), programmable logic arrays (PLAs), pro-

grammable array logic devices (PALs), or digital signal processors (DSPs). Each of the driver circuits **14** and receivers **22** may include one or more of the processors **1086**.

Processor **1086** can implement processes of various aspects described herein. Processor **1086** can be or include one or more device(s) for automatically operating on data, e.g., a central processing unit (CPU), microcontroller (MCU), desktop computer, laptop computer, mainframe computer, personal digital assistant, digital camera, cellular phone, smartphone, or any other device for processing data, managing data, or handling data, whether implemented with electrical, magnetic, optical, biological components, or otherwise. Processor **1086** can include Harvard-architecture components, modified-Harvard-architecture components, or Von-Neumann-architecture components.

The phrase "communicatively connected" includes any type of connection, wired or wireless, for communicating data between devices or processors. These devices or processors can be located in physical proximity or not. For example, subsystems such as peripheral system **1020**, user interface system **1030**, and data storage system **1040** are shown separately from the data processing system **1086** but can be stored completely or partially within the data processing system **1086**.

The peripheral system **1020** can include one or more devices configured to provide digital content records to the processor **1086**. For example, the peripheral system **1020** can include capacitive sensors **1021** or other sensors or transmitters for sending and receiving the signals described herein. The processor **1086**, upon receipt of digital content records from a device in the peripheral system **1020**, can store such digital content records in the data storage system **1040**.

The user interface system **1030** can include a mouse, a keyboard, touchscreen, another computer (connected, e.g., via a network or a null-modem cable), or any device or combination of devices from which data is input to the processor **1086**. The user interface system **1030** also can include a display device, a processor-accessible memory, or any device or combination of devices to which data is output by the processor **1086**. The user interface system **1030** and the data storage system **1040** can share a processor-accessible memory.

In various aspects, processor **1086** includes or is connected to communication interface **1015** that is coupled via network link **1016** (shown in phantom) to network **1050**. For example, communication interface **1015** can include an integrated services digital network (ISDN) terminal adapter or a modem to communicate data via a telephone line; a network interface to communicate data via a local-area network (LAN), e.g., an Ethernet LAN, or wide-area network (WAN); or a radio to communicate data via a wireless link, e.g., WiFi or GSM. Communication interface **1015** sends and receives electrical, electromagnetic or optical signals that carry digital or analog data streams representing various types of information across network link **1016** to network **1050**. Network link **1016** can be connected to network **1050** via a switch, gateway, hub, router, or other networking device.

Processor **1086** can send messages and receive data, including program code, through network **1050**, network link **1016** and communication interface **1015**. For example, a server can store requested code for an application program (e.g., a JAVA applet) on a tangible non-volatile computer-readable storage medium to which it is connected. The server can retrieve the code from the medium and transmit it through network **1050** to communication interface **1015**. The received code can be executed by processor **1086** as it is received, or stored in data storage system **1040** for later execution.

Data storage system **1040** can include or be communicatively connected with one or more processor-accessible memories configured to store information. The memories can be, e.g., within a chassis or as parts of a distributed system. The phrase "processor-accessible memory" is intended to include any data storage device to or from which processor **1086** can transfer data (using appropriate components of peripheral system **1020**), whether volatile or non-volatile; removable or fixed; electronic, magnetic, optical, chemical, mechanical, or otherwise. Exemplary processor-accessible memories include but are not limited to: registers, floppy disks, hard disks, tapes, bar codes, Compact Discs, DVDs, read-only memories (ROM), erasable programmable read-only memories (EPROM, EEPROM, or Flash), and random-access memories (RAMs). One of the processor-accessible memories in the data storage system **1040** can be a tangible non-transitory computer-readable storage medium, i.e., a non-transitory device or article of manufacture that participates in storing instructions that can be provided to processor **1086** for execution.

In an example, data storage system **1040** includes code memory **1041**, e.g., a RAM, and disk **1043**, e.g., a tangible computer-readable rotational storage device such as a hard drive. Computer program instructions are read into code memory **1041** from disk **1043**. Processor **1086** then executes one or more sequences of the computer program instructions loaded into code memory **1041**, as a result performing process steps described herein. In this way, processor **1086** carries out a computer implemented process. For example, steps of methods described herein, blocks of the flowchart illustrations or block diagrams herein, and combinations of those, can be implemented by computer program instructions. Code memory **1041** can also store data, or can store only code.

Various aspects described herein may be embodied as systems or methods. Accordingly, various aspects herein may take the form of an entirely hardware aspect, an entirely software aspect (including firmware, resident software, micro-code, etc.), or an aspect combining software and hardware aspects. These aspects can all generally be referred to herein as a "service," "circuit," "circuitry," "module," or "system."

Furthermore, various aspects herein may be embodied as computer program products including computer readable program code stored on a tangible non-transitory computer readable medium. Such a medium can be manufactured as is conventional for such articles, e.g., by pressing a CD-ROM. The program code includes computer program instructions that can be loaded into processor **1086** (and possibly also other processors), to cause functions, acts, or operational steps of various aspects herein to be performed by the processor **1086** (or other processor). Computer program code for carrying out operations for various aspects described herein may be written in any combination of one or more programming language(s), and can be loaded from disk **1043** into code memory **1041** for execution. The program code may execute, e.g., entirely on processor **1086**, partly on processor **1086** and partly on a remote computer connected to network **1050**, or entirely on the remote computer.

Various aspects are inclusive of combinations of the aspects described herein. References to "a particular aspect" (or "embodiment" or "version") and the like refer to features that are present in at least one aspect of the invention.

Separate references to "an aspect" (or "embodiment") or "particular aspects" or the like do not necessarily refer to the same aspect or aspects; however, such aspects are not mutually exclusive, unless so indicated or as are readily apparent to one of skill in the art. The use of singular or plural in referring to "method" or "methods" and the like is not limiting. The word "or" is used herein in a non-exclusive sense, unless otherwise explicitly noted.

The invention has been described in detail with particular reference to certain preferred aspects thereof, but it will be understood that variations, combinations, and modifications can be effected by a person of ordinary skill in the art within the spirit and scope of the invention.

The invention claimed is:

1. A communication interference rejection system, comprising:

a transmitter connected to a sensor, the sensor near a user's skin; and

a receiver having a capacitive termination, wherein a load impedance of the termination is a capacitance or a capacitance in parallel with a high resistance to achieve flat-band channel loss, the capacitive termination lowering loss at low frequencies as compared to a resistive termination;

wherein low frequencies in the range of 10 KHz-10 MHz are used in combination with the capacitive termination and a voltage-mode communication between the transmitter and receiver to enable simultaneous low-radiation and low channel loss.

2. The system of claim **1**, wherein the capacitive termination is implemented in combination with a capacitive return-path in human body communication to enable a square-wave like received signal from a square-wave like transmitted signal.

3. The system of claim **2**, wherein the receiver is configured to:

a. receive a signal transmitted through the body of the user, the signal comprising receive NRZ signaling or other constant amplitude signals, the receiver comprising an integrating dual data-rate receiver;

b. reject interference below a data rate by sampling the slowly changing interference;

c. provide a packet structure with short no-data periods followed by long data periods;

d. sample the slowly changing interference during the no-data period;

e. subtract the sampled interference from subsequent received signal+interference during the data period, leading to ability to reject interferences less than data-rate in broadband human body communication systems; and

f. apply the sampled interference to an input of an integrator of the integrating dual data-rate receiver to achieve the subtraction.

4. The system of claim **3**, wherein the receiver is further configured to:

g. provide Time-domain Interference-Sampling and subtraction (TD-ISS) leading to a reduction DC and low-frequency gain in transfer function of the integrating dual data-rate receiver or rejection of low-frequency interferences;

h. receive a signal transmitted through the body of the user, the signal comprising a relatively substantially small constant amplitude component and a relatively large sinusoidal or modulated interference component, said interference component due to human body antenna effect;

i. integrate the signal and sample at a sampling time;

j. sample interference just before the end of a reset period; and

k. hold the sampled value at the non-signal input of the integrator of the integrating dual data-rate receiver.

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